

ELECTROSTATICS

Electrostatics

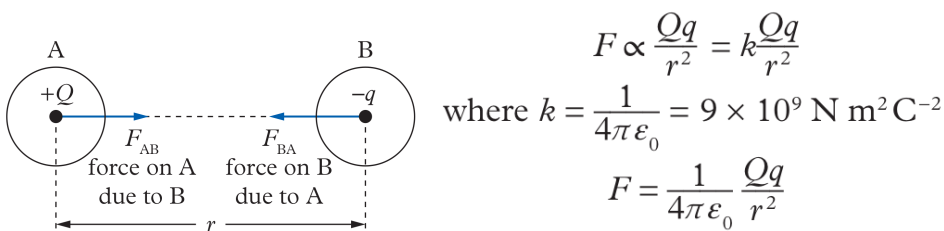
Electrostatics

- Describe Coulomb's Law.
- Solve problems using $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2} = \frac{kQq}{r^2}$.
- Describe the concepts of electric fields, electric field strength and electrical potential energy.
- Solve problems involving electric field strength using $E = \frac{F}{Q} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = \frac{kq}{r^2}$.
- Solve problems involving the work done when an electric charge is moved in an electric field using $V = \frac{\Delta U}{q}$.

distances between charges and the forces acting on the charges, Coulomb was able to show that the:

- magnitude of the force was proportional to the product of the charges
- magnitude of the force was inversely proportional to the square of the distance separating the charges
- direction of the force was along a line joining the centres of the charges
- magnitude of the force was dependent on the medium in which the charges were placed.

These points can be summarised mathematically as:



Coulomb's law

states that like electric charges repel and opposite electric charges attract, with a force proportional to the product of the electric charges and inversely proportional to the square of the distance between them, expressed by the formula $F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2}$



- Describe Coulomb's Law.

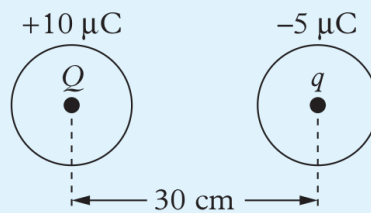
$$F = k \frac{Qq}{r^2}$$

Forces in a line

The simplest problems involving Coulomb's law are for objects in a straight line. This can be two charged objects such as two protons in a nucleus, or three or more charged objects in a straight line. The vector sum for linear arrangements is much simpler than for two-dimensional set-ups.

WORKED EXAMPLE 6.1A

Consider the system of two charges as shown in Figure 5. If $Q = +10 \mu\text{C}$ and $q = -5 \mu\text{C}$, and the separation distance is 30 cm, calculate the force between them.



$$1 \mu\text{C} = 10^{-6} \text{C}$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2} = k \frac{Qq}{r^2} = 9 \times 10^9 \times \frac{10 \times 10^{-6} \times (+5) \times 10^{-6}}{0.3^2}$$

Signs are omitted.

$$F = 5 \text{ N}$$

The charges are unlike, so $F = 5 \text{ N}$ (attraction)

Physics 2025



Electrical circuits

$I = \frac{q}{t}$	$P = I^2 R$
$V = \frac{W}{q}$	$V_t = V_1 + V_2 + \dots V_n$
$P = \frac{W}{t}$	$R_t = R_1 + R_2 + \dots R_n$
$V = IR$	$I_t = I_1 + I_2 + \dots I_n$
$P = VI$	$\frac{1}{R_t} = \frac{1}{R_1} + \frac{1}{R_2} + \dots \frac{1}{R_n}$

Electromagnetism

$F = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2} = \frac{kQq}{r^2}$	$F = qvB \sin \theta$
$E = \frac{F}{q} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = \frac{kq}{r^2}$	$\phi = BA \cos \theta$
$V = \frac{\Delta U}{q}$	$emf = -\frac{N\Delta(BA_{\perp})}{\Delta t}$
$B = \frac{\mu_0 I}{2\pi r}$	$emf = -N \frac{\Delta \phi}{\Delta t}$
$B = \mu_0 n I$	$I_p V_p = I_s V_s$
$F = BIL \sin \theta$	$\frac{V_p}{V_s} = \frac{N_p}{N_s}$

Physical constants and unit conversions

Heating processes	
Specific heat capacity of ice	$c_i = 2.05 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$
Specific heat capacity of water	$c_w = 4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$
Specific heat capacity of steam	$c_s = 2.00 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$
Latent heat of fusion for water	$L_f = 3.34 \times 10^5 \text{ J kg}^{-1}$
Latent heat of vaporisation for water	$L_v = 2.26 \times 10^6 \text{ J kg}^{-1}$

Ionising radiation and nuclear reactions	
Electron volt	$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$
Speed of light in a vacuum	$c = 3 \times 10^8 \text{ m s}^{-1}$
Atomic mass unit	$1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$
Mass of an alpha particle	$m_\alpha = 6.6446572 \times 10^{-27} \text{ kg}$
Mass of an electron	$m_e = 9.1093835 \times 10^{-31} \text{ kg}$
Mass of a neutron	$m_n = 1.6749275 \times 10^{-27} \text{ kg}$
Mass of a proton	$m_p = 1.6726219 \times 10^{-27} \text{ kg}$

Electrical circuits	
Charge on an electron	$e = -1.6 \times 10^{-19} \text{ C}$

Electromagnetism	
Coulomb's constant	$\frac{1}{4\pi\epsilon_0} = k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Magnetic permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ T m A}^{-1}$

Linear motion and force	
Mean acceleration due to gravity on Earth	$g = 9.8 \text{ m s}^{-2}$

Labelling forces

Labelling the forces can be tricky, but it is good to be consistent. Consider a pair of charged objects A and B (Figure 6). Each experiences a force due to the other. The force on A due to B is labelled as F_{AB} . It represents $F_{\text{on A due to B}}$. Using this order for the subscripts helps you keep track of the various forces and what objects are being considered. It is important to recognise that both objects are responsible for the force (agent and receiver) and the magnitudes are equal.

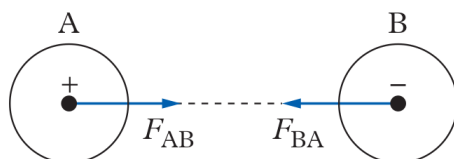


FIGURE 6 Labelling the forces

WORKED EXAMPLE 6.1B

A system of three charges in a line consists of object A with $Q_A = +10 \mu\text{C}$, $Q_B = -10 \mu\text{C}$, and $Q_C = +10 \mu\text{C}$, separated by distances to their centres as shown in Figure 7. Distances are measured to 2 significant figures.

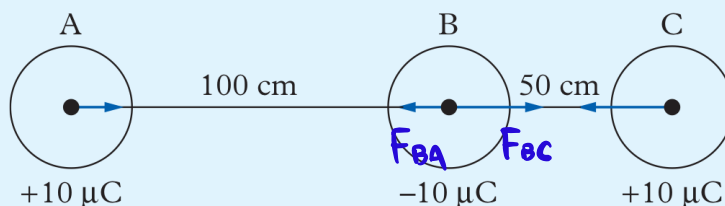


FIGURE 7 Multiple charges in one dimension

Calculate the net force on B due to A and C.

F_{BA} & F_{BC}

① F_{BA}

$$F_{BA} = k \frac{Q_A Q_B}{r_{AB}^2} = 9 \times 10^9 \times \frac{10 \times 10^{-6} \times 10 \times 10^{-6}}{1^2}$$

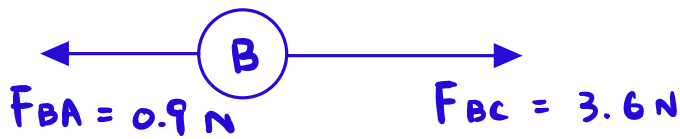
= 0.9 N attraction to the left

② F_{BC}

$$F_{BC} = k \frac{Q_B Q_C}{r_{BC}^2} = 9 \times 10^9 \times \frac{10 \times 10^{-6} \times 10 \times 10^{-6}}{0.5^2}$$

= 3.6 N attraction, to the right

③ Net force on B



$$F_{NET} = 3.6 - 0.9 = 2.7 \text{ N to the right.}$$

Forces in two-dimensions

For three or more objects not in a line, the vector sum becomes more difficult.

WORKED EXAMPLE 6.1C

Consider a system of three charges $Q_A = +5 \mu\text{C}$, $Q_B = +5 \mu\text{C}$ and $Q_C = -5 \mu\text{C}$, and separation distances as shown in Figure 8. The lines between the charges make a right-angled triangle with interior angles of 45° . Calculate the force acting on charge object Q_C .

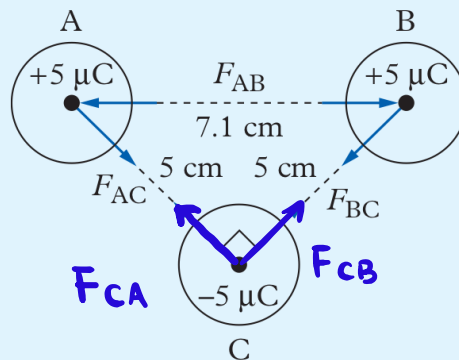


FIGURE 8 Charges in two dimensions

① F_{CA} .

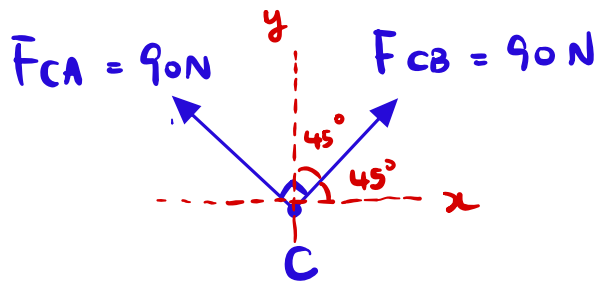
$$\begin{aligned} F_{CA} &= k \frac{Q_C Q_A}{r_{CA}^2} = 9 \times 10^9 \times \frac{5 \times 10^{-6} \times 5 \times 10^{-6}}{0.05^2} \\ &= 90 \text{ N attraction} \end{aligned}$$

② F_{CB}

$$F_{CB} = k \frac{Q_C Q_B}{r_{CB}^2} = 9 \times 10^9 \times \frac{5 \times 10^{-6} \times 5 \times 10^{-6}}{0.05^2}$$

$$= 90 \text{ N attraction}$$

③ Net force on C ($\uparrow$)



$$F_{CB,x} = 90 \cos 45^\circ = 45\sqrt{2} \text{ N}$$

$$F_{CB,y} = 45\sqrt{2} \text{ N}$$

$$F_{CA,x} = -45\sqrt{2} \text{ N}$$

$$F_{CA,y} = 45\sqrt{2} \text{ N}$$

$$F_x = F_{CB,x} + F_{CA,x} = 0$$

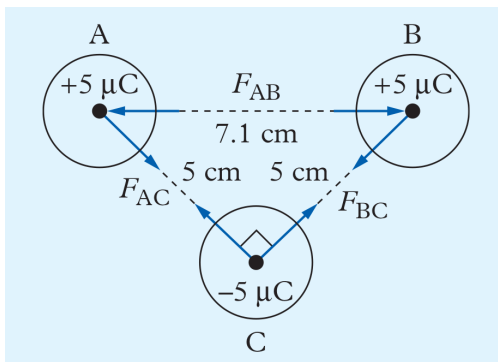
$$F_y = F_{CB,y} + F_{CA,y} = 45\sqrt{2} + 45\sqrt{2} = 90\sqrt{2} \text{ N} \uparrow$$

$$= 127 \text{ N}$$

$$F = \sqrt{F_x^2 + F_y^2} = 127 \text{ N} \quad (\text{Vertically upward})$$

HW.

calculate the net force on object B.



1 **Define** 'Coulomb's law' and state the numerical value of the constant to 3 significant figures, including units.

3 **Calculate** the size of the Coulomb repulsion force between two protons in a helium nucleus, given that they are separated by a distance of 1×10^{-14} m. Recall that the magnitude of the charge on a proton is the elementary charge.

5 Two point charges A and B each with a charge $+Q$ coulombs are separated by a distance of r metres. The force acting between them is 6×10^{-4} N. **Determine:**

- a whether the force is attractive or repulsive
- b the magnitude of the force if distance r is doubled
- c the magnitude of the force if charge Q on both A and B is doubled
- d the magnitude of the force if charge Q on both is halved, and so is the distance between them.

6 Consider the system of charges in Figure 12.

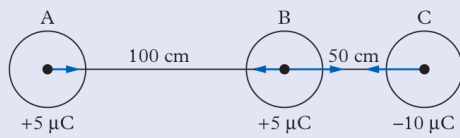


FIGURE 12 Three charges in one dimension

Determine the net force on:

- a B
- b C

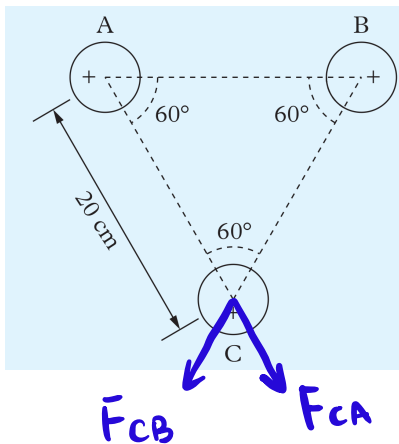
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$$F = k \frac{q_1 q_2}{r_{12}^2}$$

$$k = 9 \times 10^9 \text{ N} \cdot \text{m}^2 \cdot \text{C}^{-2}$$

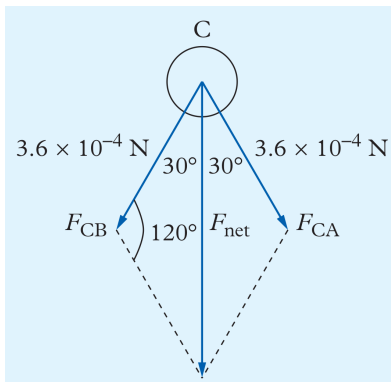
WORKED EXAMPLE 6.1E

Consider the system of charges in Figure 11. Deduce the magnitude and direction of the force on charge C. Each charge is $+4.0 \times 10^{-8} \text{ C}$ and they are arranged in an equilateral triangle of side 20 cm (2.0×10^{-1}).



$$F_{CB} = k \frac{q_B q_C}{r_{BC}^2} = 9 \times 10^9 \frac{4 \times 10^{-8} \times 4 \times 10^{-8}}{0.2^2}$$
$$= 3.6 \times 10^{-4} \text{ N}$$

$$F_{CA} = 3.6 \times 10^{-4} \text{ N}$$



Cosine Rule

$$F^2 = F_{CA}^2 + F_{CB}^2 - 2 \times F_{CA} \times F_{CB} \cos 120^\circ$$
$$= 3.89 \times 10^{-7} \text{ N}$$

$$F_{\text{NET}} = 6.2 \times 10^{-4} \text{ N}$$

The force is repulsive and will be directed vertically down.

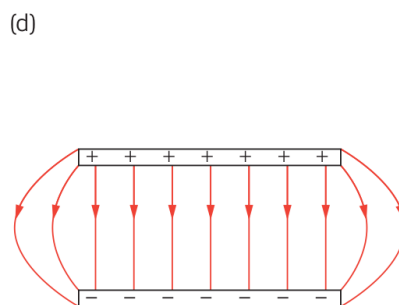
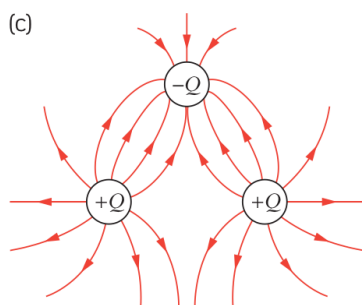
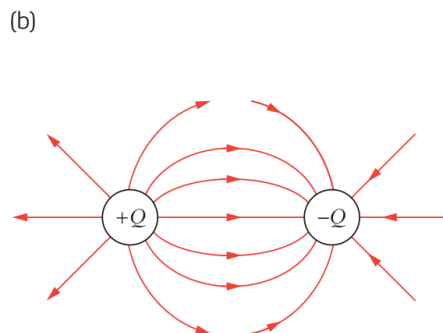
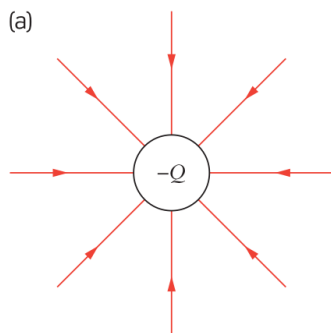
Electric fields and field strength

Electrostatics

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- Solve problems involving the work done when an electric charge is moved in an electric field using $V = \frac{\Delta U}{q}$.

Electric field : A region of space near an electrically charged particle or object within which a force would be exerted on other electrically charged particles or objects.

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Electric field strength E [N C^{-1}]

The intensity of an electric field at a particular location.

We can have strong electric fields, such as a bolt of lightning, or weak electric fields, such as when you rub a plastic comb on a piece of wool. As force is a vector quantity, **electric field strength** is also a vector quantity. It has both magnitude and direction.

The main difference between electric fields and gravitational fields is that gravitational fields have only attractive forces, whereas electric fields can provide both attractive and repulsive forces. The intensity of the electric field – its electric field strength – at any point is defined as the force acting on a test unit charge, q , placed at that point in the field. The direction of the electric field at any point is given by the resultant force direction acting on a positive test charge placed at that point in the field. Thus mathematically, electric field strength is defined by the following equation:

$$\vec{E} = \frac{\vec{F}}{q}$$

where E is the electric field strength in units of N C^{-1} (newton per coulomb), F (N) is the force acting on a charge in the field, and q is the size of the test charge in coulomb (C). Refer to Figure 4.

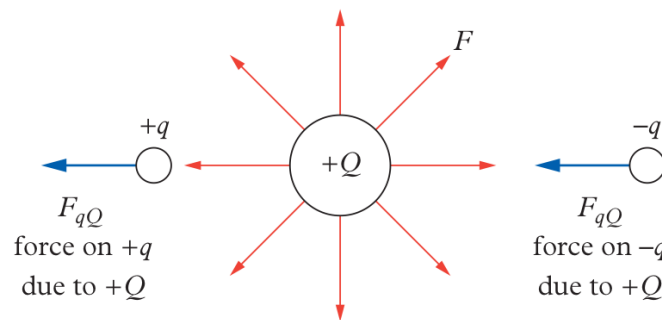


FIGURE 4 An electric field surrounds $+Q$ and test charges ($+q$ and $-q$) experience forces in the field.

WORKED EXAMPLE 6.2A

A short length of nichrome wire is held vertically under an overhead power wire where the field strength is 5000 N C^{-1} downwards. Calculate the force experienced by an electron in the wire, $q_e = -1.60 \times 10^{-19} \text{ C}$.

$$E = 5000 \text{ N C}^{-1}$$

$$q_e = -1.6 \times 10^{-19} \text{ C}$$

$$E = \frac{F}{q} \Rightarrow F = qE$$

$$= -1.6 \times 10^{-19} \times 5000$$

$$= 8 \times 10^{-16} \text{ N}$$

The field is directed downwards so the force on a negative charge acts upwards.

Uniform electric fields

Notice that in Figure 5 the electric field between parallel charged plates is very uniform in nature. This **uniform electric field** will produce a constant force on any test charge held between the plates, independent of position. In 1909, the American physicist Robert Millikan used the very uniform electric field between charged parallel metallic plates to investigate the nature of electric charge itself. Millikan was able to balance the electric force on charged oil droplets sprayed between the plates with the droplets' own gravitational weight. By carefully measuring the mass of these oil droplets and changing their electric charge with X-rays, Millikan was able to calculate a value for the charge on an electron. He won the Nobel Prize in Physics in 1923 for his work on the electrostatics of elementary charges.

uniform electric field

a field that has constant field strength, as found between charged parallel plates

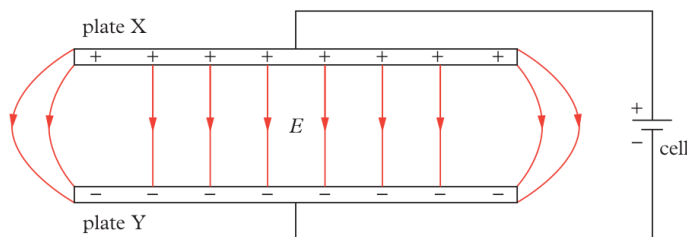


FIGURE 5 A uniform electric field between oppositely charged parallel plates

Newton's laws of motion and electric fields

The electric field strength has been defined as the force acting on a charged particle in the field. You saw that if the field strength and the charge on a particle are known, the force acting can be calculated.

Newton's laws of motion describe the effect of forces acting on matter. You have already considered Newton's third law, so now consider the first and second laws in turn to assess how they relate to electrostatic forces.

Newton's first law

A charged particle, Q , is fired at constant speed into a uniform electric field between two charged plates (Figure 8).

The direction of the electric field vector is pointing down from the positive plate to the negative plate, as shown by the arrows. This means that the direction of a positively charged particle in the field is also down.

Note: The field strength vector and force vector on a positive test charge are both in the same direction.

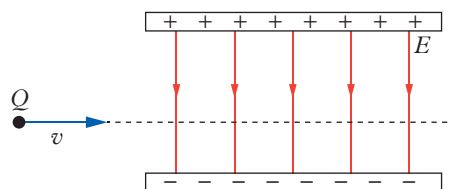


FIGURE 8 Charged particle in a uniform field

If a positive particle is fired into the field at constant speed from the left, it would continue at a constant velocity unless acted on by an outside force (Newton's first law). In this case, there is a force and it will cause a positive particle to move downwards, thereby changing its velocity. The bigger the force, the greater the deflection away from the horizontal path. If the particle was negative it would move upwards.

Newton's second law

The magnitude of the force can be determined by using Newton's second law:

$$\vec{a} = \frac{\vec{F}}{m}$$

This is illustrated in Worked example 6.2B.



FIGURE 7 Electric eels can generate an electric field of up to 800 N C^{-1} in order to stun prey.

WORKED EXAMPLE 6.2B

A proton ($q = +1.60 \times 10^{-19} \text{ C}$, $m = 1.67 \times 10^{-27} \text{ kg}$) is introduced at rest into the electric field of a mass spectrometer that has a constant electric field strength of $4.00 \times 10^3 \text{ N C}^{-1}$, as shown in Figure 9.

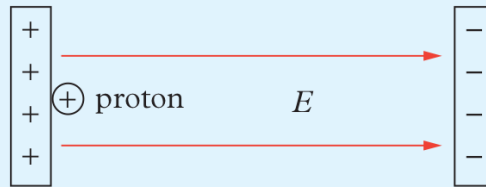


FIGURE 9 Electric field in a mass spectrometer

Calculate:

- a the force acting on the proton due to the electric field
- b the acceleration of the proton in the direction of the field
- c the time for the proton to reach a speed of $2.00 \times 10^5 \text{ m s}^{-1}$
- d the distance travelled in going from rest to this speed.

$$\begin{aligned} \text{(a)} \quad F &= qE \\ &= 1.6 \times 10^{-19} \times 4 \times 10^3 \\ &= 6.4 \times 10^{-16} \text{ N} \quad (\text{To the right}) \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad a &= \frac{F}{m} \\ &= \frac{6.4 \times 10^{-16}}{1.67 \times 10^{-27}} \\ &= 3.83 \times 10^{11} \text{ m/s}^2 \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad v &= 2 \times 10^5 \text{ m/s} & a &= 3.83 \times 10^{11} \text{ m/s}^2 \\ u &= 0 \text{ m/s} \end{aligned}$$

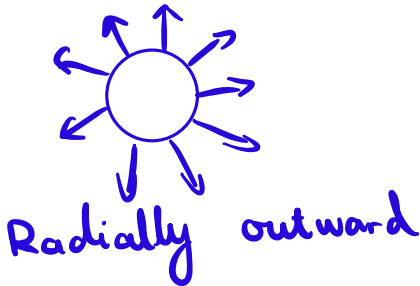
$$v = u + at \Rightarrow t = \frac{v - u}{a} = \frac{2 \times 10^5}{3.83 \times 10^{11}} = 5.22 \times 10^{-7} \text{ s}$$

$$\text{(d)} \quad s = \cancel{v^2} + \frac{1}{2}at^2 = \frac{1}{2} \times 3.83 \times 10^{11} \times (5.22 \times 10^{-7})^2 = 0.0522 \text{ m}$$

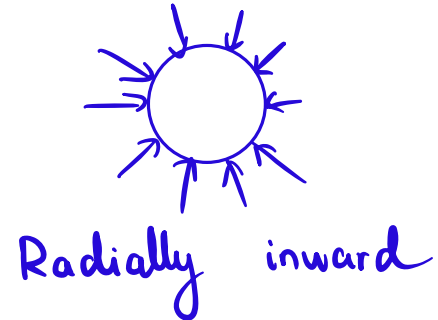
Electric field strength formula

Consider a point source of positive charge, $+Q$. If a small positive test charge, $+q$, was near the charge $+Q$, then forces of repulsion would be felt in radial lines directed outwards. The further away $+q$ was moved along these lines the less intense the electric field strength would be. At all times the magnitude of the force would be given by Coulomb's law and thus the electric field strength, E , at some distance, r , from the charge, $+Q$, would be:

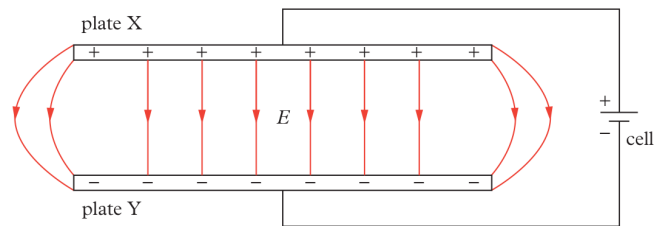
$$\begin{aligned}\vec{E} &= \frac{\vec{F}}{q} \\ &= F \times \frac{1}{q} \\ &= \frac{kQq}{r^2} \times \frac{1}{q} \\ E &= \frac{kQ}{r^2}\end{aligned}$$



and its direction is radially outwards.



This equation therefore describes the electric field strength at a point r from a large point charge $+Q$. The electric field strength E is defined as having a value of zero at an infinite distance from the source. Note that diagrams such as that in Figure 5 on page 164 represent the nature of the electric field in a two-dimensional cross-section only and the actual zone of influence is always in three dimensions around point charges.



WORKED EXAMPLE 6.2C

Calculate the electric field strength 1 nm (1.0×10^{-9} m) from a proton ($q_p = +1.60 \times 10^{-19}$ C).

$$\begin{aligned}E &= \frac{F}{q} = \frac{kq_p}{r^2} = \frac{9 \times 10^9 \times 1.6 \times 10^{-19}}{(1 \times 10^{-9})^2} \\ &= 1.4 \times 10^9 \text{ N/C} \\ &\text{(Directed away from the proton)}\end{aligned}$$

WORKED EXAMPLE 6.2D

Figure 10 shows two point charges A and B separated by a distance of 15.0 cm.

- Determine the magnitude and direction of the electric field at point X midway between the two charges.
- Determine the point between the charges at which the electric field strength is zero.

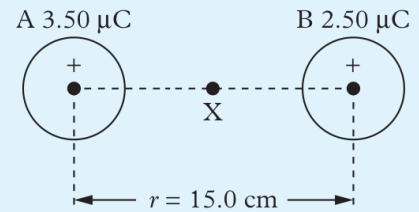


FIGURE 10

(a) E_A , E_B , E_{net}

$$E_A = \frac{kQ_A}{r^2} = \frac{9 \times 10^9 \times 3.5 \times 10^{-6}}{(0.075)^2} = 5.59 \times 10^6 \text{ N/C}$$

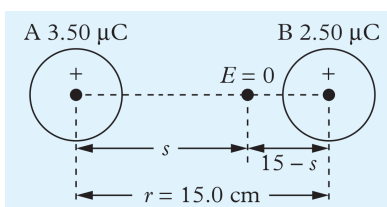
(to the right)

$$E_B = \frac{kQ_B}{r^2} = \frac{9 \times 10^9 \times 2.5 \times 10^{-6}}{0.075^2} = 4 \times 10^6 \text{ N/C} \quad (\leftarrow)$$

$$E_{\text{net}} = (5.59 - 4) \times 10^6 = 1.59 \times 10^6 \text{ N/C} \quad (\text{To the right, towards B})$$

$$\begin{array}{c} \leftarrow \cdot \rightarrow 5.59 \times 10^6 \text{ N/C} \\ 4 \times 10^6 \text{ N/C } \times \end{array}$$

(b) The point at which the electric field is zero is the point where the electric field $|E_A| = |E_B|$.



$$E_A = \frac{kQ_A}{s^2}$$

$$E_B = \frac{kQ_B}{(15 - s)^2}$$

$$E_A = E_B$$

$$\frac{\cancel{9 \times 10^9} \times 3.5 \times \cancel{10^{-6}}}{S^2} = \frac{\cancel{9 \times 10^9} \times 2.5 \times \cancel{10^{-6}}}{(15-S)^2}$$

$$\frac{3.5}{S^2} = \frac{2.5}{(15-S)^2}$$

$$\left(\frac{15-S}{S}\right)^2 = \frac{2.5}{3.5}$$

$$\frac{15-S}{S} = \sqrt{\frac{2.5}{3.5}}$$

$$\frac{15-S}{S} = 0.8452$$

$$15-S = 0.8452S$$

$$S = 8.13 \text{ cm from A}$$

Electric potential and energy

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- ✓ Solve problems involving electric field strength using $E = \frac{F}{Q} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = \frac{kq}{r^2}$.
- Solve problems involving the work done when an electric charge is moved in an electric field using $V = \frac{\Delta U}{q}$.

Electric potential energy U [J]

The capacity of electric charge carrier to do work due to their position in an electric field.

Electrical potential energy

The gravitational field analogy can be used again when discussing the energy stored in an electric field. When a mass moves in the direction of the gravitational field (towards the ground) it goes from high gravitational potential energy to low. That is, it goes from a region of high potential to a region of lower potential. Similarly, when a positive charge moves in the direction of an electric field it too goes from high potential energy to lower potential energy.

Conversely, work was needed to lift a mass against the direction of the gravitational

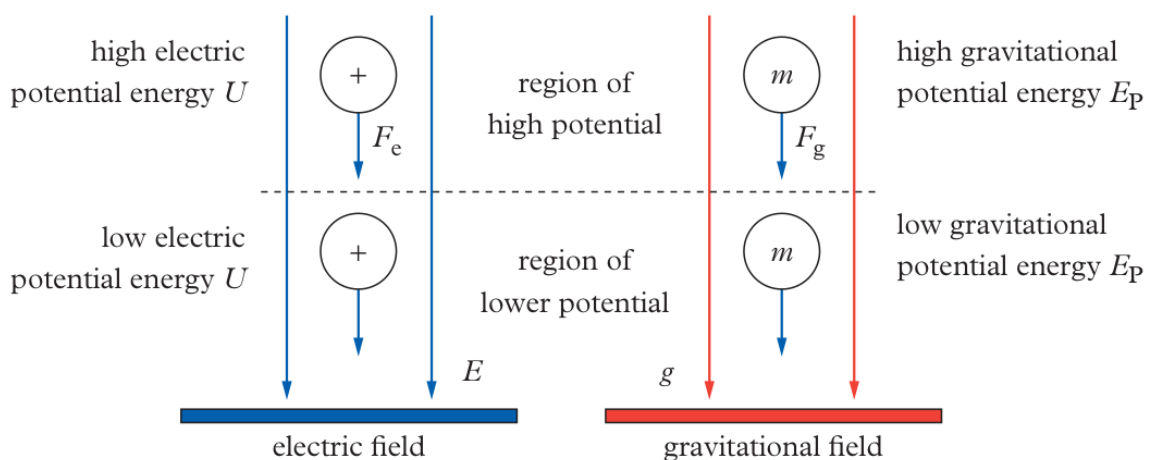


FIGURE 1 A gravitational field analogy for electric fields

field and, as this work was done, gravitational potential energy was stored in the combined masses and the gravitational field. The energy stored was equivalent to the work done, W , and calculated using $W = mgh$. Similarly, it requires work to be done to move a positive electric charge against the direction of the field. The charge and the field acquire **electrical potential energy** (U) by the work done. For an electric field, the work done is given by $W = qEd$. You should be able to see the similarity in Table 1.

TABLE 1 Gravitational and electrical quantities compared

Quantity	Gravitational field	Electric field
Work done	work, W (in J) $W = mgh$	work, W (in J) $W = qEd$
Property of the objects	mass, m (in kg)	charge, q (in C)
Field strength	gravitational, g (N kg ⁻¹)	electric, E (in N C ⁻¹)
Distance moved	height, h (in m)	distance, d (in m)

If the charge is moved perpendicular to the field lines then no work at all will be done .

The work done in moving charge $+q$ will subsequently store electrostatic potential energy in the charge $+q$. If the repulsive force between the charges is allowed to act by itself, then the stored electrical potential energy will be converted back into kinetic energy of motion as the particles fly away from each other. This effect can be used in an electrostatic charge accelerator such as the cyclotron, which produces a beam of charged particles that can be used for medical, industrial and research processes.

Electric potential

Calculating the work done is difficult as the field is not uniform. It is better to have a physical quantity that is independent of the test charge, which is defined as **electric potential** (V). It is a point in space within any given electric field. That is, it is not the energy of a system of charges but merely a point in space.

The electric potential, V , at that point would be the work done if you moved a unit positive charge (+1 C) from infinity to that point. At infinity, the electric potential and electrical potential energy are defined as being zero ($V_{\infty} = 0$, $U_{\infty} = 0$), so ΔV and ΔU are the same as V and U respectively. This is shown in Figure 2.

electric potential
the electrostatic potential energy stored per unit charge at any given point (symbol: V ; SI unit: volt; unit symbol: V)

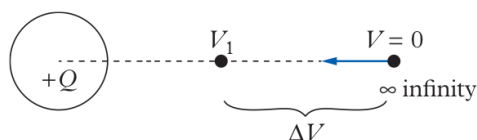


FIGURE 2 Electric potential changes when a charge is brought in from infinity.

Hopefully, you can see the difference: electrical potential energy is the actual energy residing in a charged object and the field at that point; the electric potential is just a measure of the energy per coulomb of an actual charged object at that point.

The electric potential, V , is thus the electrical potential energy stored per unit charge at any given point. Thus:

$$\Delta V = \frac{\Delta U}{q}$$

↗ electrical potential energy
↘ charges.

Electrical potential difference

Electric potential at a point is a scalar quantity and is measured in units of joules per coulomb (J C^{-1}). One joule per coulomb is also known as a volt: 1 volt (V) = 1 J C^{-1} . Of course, moving unit positive charges from infinity to a point within the field of a point charge Q is not particularly realistic, and hence the term **electrical potential difference** is more useful, in that it describes the difference in electric potential between two positions within the field (V_1 to V_2), neither of which is at infinity (Figure 3).

$$\Delta V = V_2 - V_1 = \frac{\Delta U}{q}$$

The potential difference ΔV is often given the symbol V , the same unit as potential. This is not strictly correct, but it is used for convenience. For example, a 9 V battery isn't labelled as a potential difference of 9 V, it just says 9 V and it is assumed that the negative terminal is zero volts. The potential difference formula is usually written as:

$$V = \frac{\Delta U}{q}$$

but should be interpreted as:

$$\Delta V = \frac{\Delta U}{q}$$

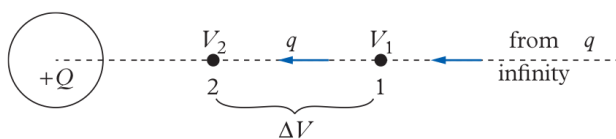


FIGURE 3 Electrical potential difference between two points in a field

electrical potential difference

the work done in moving a unit charge between the final and the initial positions in an electric field (symbol: ΔV ; SI unit: volt; unit symbol: V)

WORKED EXAMPLE 6.3A

Calculate the potential difference when $\Delta U = 15 \text{ J}$ is used to move $q = 2.5 \text{ C}$ through an electric field in a wire.

$$V = \frac{\Delta U}{q} = \frac{15}{2.5} = 6 \text{ V}$$

The potential difference is 6 V

Equipotential lines

Points in space that all have the same potential make up what is called an equipotential line, or, if they are in three dimensions, an equipotential surface. A series of equipotential lines with each line representing a different value of potential can be used to represent the electric field. Figure 4 is an example of this.

No net work is done on a charge by an electric field when the charge moves along the equipotential lines. This is a consequence of $V_f - V_i = \frac{\Delta U}{q}$. That is, if there is no potential difference, no work is done when a charge is moved from one point to another. However, work is done when shifting from one equipotential line to another.

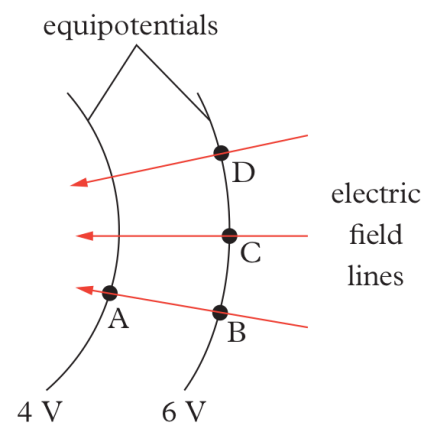


FIGURE 4 No work is done along an equipotential line (B, C, D).

WORKED EXAMPLE 6.3B

In Figure 4, what are the differences in electric potential for:

a $V_B - V_A$

b $V_C - V_A$

c $V_C - V_B$

SOLUTION

a $\Delta V = V_B - V_A$
 $= 6 - 4$
 $= 2 \text{ V}$

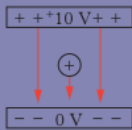
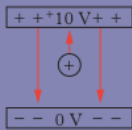
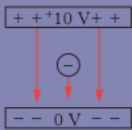
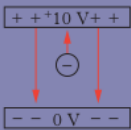
b $\Delta V = V_C - V_A$
 $= 6 - 4$
 $= 2 \text{ V}$

c $\Delta V = V_C - V_B$
 $= 6 - 6$
 $= 0 \text{ V}$

Transfers and transformations of energy

When a charged object moves in a field, it can either gain or lose potential energy if there is a net force acting on it. This change is associated with a change in electric potential. The following description details what happens when a positive or negative charge moves with or against a field. That means there are four possible scenarios and the changes are detailed in Table 2.

TABLE 2 Energy changes in moving charges in an electric field

	CASE A positive charge moving with the field 	CASE B positive charge moving against the field 	CASE C negative charge moving with the field 	CASE D negative charge moving against the field 
Motion of charge in field	with (down)	against (up)	with (down)	against (up)
Force on charge, F_{net}	down	down	up	up
Acceleration, a	down (speeds up)	down (slows down)	up (slows down)	up (speeds up)
Kinetic energy, ΔE_k	increases (+)	decreases (-)	decreases (-)	increases (+)
Potential energy, ΔU	decreases (ΔU is -)	increases (ΔU is +)	increases (ΔU is +)	decreases (ΔU is -)
Potential difference, $\Delta V = V_f - V_i$	$0 - +10 = -10 \text{ V}$ decreases	$+10 - 0 = +10 \text{ V}$ increases	$0 - +10 = -10 \text{ V}$ decreases	$+10 - 0 = +10 \text{ V}$ increases
Work done	by the field	on the field	on the field	by the field

- Case A:** When a positively charged object enters an electric field, it will experience a net force in the direction of the field (by definition). Therefore, it will accelerate in that direction. This means its kinetic energy (E_k) will change. If the force is in the direction of motion, as in Case A, the object will speed up and its kinetic energy will increase (ΔE_k is positive). By the law of conservation of energy, the gain in kinetic must be offset by a loss in potential energy (U); therefore, U decreases and thus ΔU ($U_{\text{final}} - U_{\text{initial}}$) is negative. The field (U) has lost energy, so work has been done *by* the field (on the object).
- Case B:** When the positively charged object enters the electric field it experiences a net force downwards in the direction of the field (again, by definition). Therefore, the direction of the acceleration will be down, but as it is moving upwards it will slow down and its kinetic energy will decrease (ΔE_k is negative). The loss in kinetic energy is offset by a gain in potential energy (U), therefore U increases and thus ΔU is positive. The field (U) has gained energy so work is done *on* the field.
- Case C:** When a negatively charged object enters the electric field it experiences a net force upwards in the direction opposite to the field (by definition). Therefore, the direction of acceleration will be up, and as it is moving downwards it will slow down and its kinetic energy will decrease (ΔE_k is negative). The loss in kinetic energy means a gain in potential energy (U), therefore ΔU is positive. The field gains energy so work is done *on* the field.
- Case D:** When a negatively charged object enters the electric field it experiences a net force upwards in the direction opposite to the field (by definition). Therefore, the direction of the acceleration is up, and as it is moving upwards it will speed up and its kinetic energy will increase (ΔE_k is positive). The gain in kinetic energy is offset by a loss in potential energy (U), therefore ΔU is negative. The field loses energy so work is done *by* the field.

CHECK YOUR LEARNING 6.2

Describe and explain

- 1 **Sketch** the cross-sectional electric field diagram for a system of three negative charges situated at the corners of an equilateral triangle.
- 2 **Calculate** the magnitude and direction of the electric field strength 0.2 m from a point charge of $-6 \mu\text{C}$.

Apply, analyse and interpret

- 3 **Distinguish** between electric field and electric field strength.
- 4 Two charged spheres are placed 2.0 m apart as shown in Figure 12.

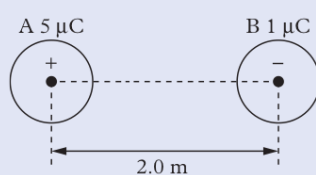


FIGURE 12 Two charges in a line

Determine:

- a the electric field strength midway between the two charges
 - b the electric field strength at a point 50 cm to the right of object B
 - c the location of a point P (besides infinity) at which the field strength is zero.
- 5 An electron ($q_e = -1.60 \times 10^{-19} \text{ C}$, $m_e = 9.109 \times 10^{-31} \text{ kg}$) is introduced at rest into an electric field with a constant electric field strength of 50 N C^{-1} .

Determine:

- a the force acting on the electron due to the electric field
- b the acceleration of the electron in the direction of the field
- c the time to reach 80% of the speed of light (use $c = 3 \times 10^8 \text{ m s}^{-1}$)
- d how far it moved in the time taken to go from rest to the final speed.

CHECK YOUR LEARNING 6.3

Describe and explain

- 1 **Describe** what it means to say that when work is done on a field, U increases.
- 2 **Explain** whether an electron entering an electric field naturally wants to travel with the field or against it.
- 3 **Calculate** the work done by an electron ($q_e = -1.60 \times 10^{-19} \text{ C}$, $m_e = 9.109 \times 10^{-31} \text{ kg}$) in travelling from the +6 V end to the 0 V end of a piece of nichrome wire 20 cm long and connected across the terminals of a 6 V battery.

Apply, analyse and interpret

- 4 **a Calculate** the kinetic energy of an electron as it enters an electric field at a speed of $0.6c$ (where $c = \text{the speed of light} = 3 \times 10^8 \text{ m s}^{-1}$).
- b Determine** the potential difference the electron will experience when it is brought to rest by transferring all of its kinetic energy to the field.
- 5 Two points in space are at electric potentials of +18 V and -6 V respectively.
 - a **Calculate** the potential difference between these two points.
 - b **Determine** the work done in moving a charge of $5.5 \mu\text{C}$ from one point to the other.

- 6 **Determine** the energy of a lightning flash when there is a potential difference of $2 \times 10^9 \text{ V}$ between the cloud and the ground, and 35 C of charge is transferred during the strike.
- 7 **Determine** the electric potential in an electric field at the points listed below. Note that Figure 5 shows two equipotential lines about a charged object. It takes 0.2 J of work to be done on a +0.1 C charge at point C to move it to point E.
 - a point E
 - b point D
 - c point B

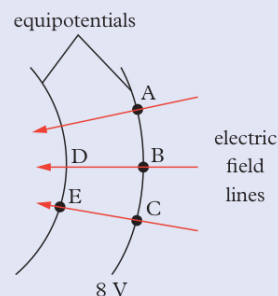


FIGURE 5

Investigate, evaluate and communicate

- 8 **Evaluate** whether the statement 'electric field lines never cross' is true.